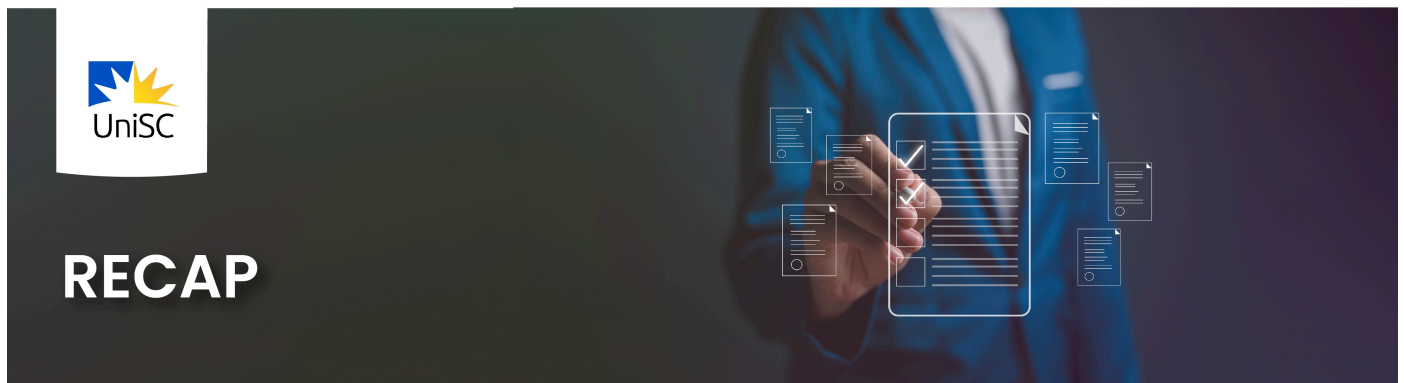


Module 3 Summary



Module 3 Review, Reflect, Consolidate, and Practice

Module 3 focused on portfolio evaluation, risk-adjusted performance, inflation-adjusted returns, and downside risk analysis. Throughout the module, you learned that evaluating investments requires much more than simply measuring returns. Effective portfolio evaluation connects returns to risk, inflation, diversification, market exposure, and the investor's objectives.

This section summarises what you have learned and links it explicitly to the course learning outcomes.

What You Learned in Module 3

Portfolio Evaluation Requires More Than Measuring Returns

You learned that portfolio evaluation involves assessing:

- Total return
- Risk exposure
- Risk-adjusted performance
- Inflation-adjusted performance
- Tail risk and downside risk
- Portfolio efficiency

You learned that:

- High returns alone do not necessarily indicate strong performance.
- Returns must be evaluated relative to the level and type of risk taken.
- Inflation can erode real purchasing power even when nominal returns appear strong.

This directly supports:

CLO1: Explain key concepts and theoretical foundations of financial analytics and their role in strategic decision-making.

You can now explain why professional portfolio evaluation requires both return and risk analysis.

Measuring Portfolio Returns

You learned how to calculate and interpret:

- Holding Period Return (HPR)
- Arithmetic average returns
- Geometric average returns

You learned that:

- HPR measures total payoff over a holding period.
- Arithmetic averages can overstate long-run performance.
- Geometric averages correctly account for compounding and volatility drag.

You now understand that:

- Long-term wealth accumulation depends on compounded growth.
- Losses have asymmetric effects because large losses require even larger gains to recover.

This supports:

CLO2: Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.

You can now calculate and interpret different return measures appropriately.

Measuring Portfolio Risk

You learned that risk represents uncertainty about returns and that different forms of risk matter in different contexts.

You distinguished between:

- Total risk
- Systematic risk
- Unsystematic risk

You learned that:

- Diversification reduces unsystematic risk.
- Systematic risk cannot be diversified away.

- Beta measures sensitivity to market movements.

You also learned that:

- Standard deviation measures total volatility.
- Risk and return must be evaluated together rather than separately.

This supports:

CLO2: Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.

You have now applied quantitative risk measures directly to investment evaluation.

Risk-Adjusted Performance Measures

You learned how to evaluate whether returns adequately compensated investors for the risk taken.

You calculated and interpreted:

- Sharpe Ratio
- Treynor Ratio

You learned that:

- The Sharpe Ratio evaluates excess return per unit of total volatility.
- The Treynor Ratio evaluates excess return per unit of systematic market risk.
- Different measures are appropriate for different portfolio structures.

You now understand that:

- A portfolio with lower raw returns can still be superior if it delivers higher risk-adjusted performance.
- Portfolio efficiency matters as much as absolute return.

This supports:

CLO3: Use business analytics tools to develop, evaluate, and communicate predictive insights that support financial problem-solving.

You can now compare portfolios using formal financial performance metrics.

Inflation and Real Returns

You learned how inflation affects investment performance through the Fisher relationship.

You learned that:

- Nominal returns can overstate real wealth creation.
- Real returns measure changes in purchasing power.
- Inflation adjustment is especially important for long-term investors and retirees.

You calculated:

- Real returns using the Fisher equation

You now understand that:

- Strong nominal returns may still represent weak real performance during periods of high inflation.

This supports:

CLO1: Explain key concepts and theoretical foundations of financial analytics and their role in strategic decision-making.

You can now distinguish between nominal and real financial performance.

Downside Risk

You explored:

- Downside Deviation
- Sortino Ratio

You learned that:

- Investors care more about downside risk than upside volatility.
- Standard deviation treats gains and losses equally, even though only losses concern investors.
- Downside deviation isolates variability below a minimum acceptable return (MAR).

You now understand that:

- Portfolios with the same average return can carry very different downside risk.
- The Sortino ratio measures excess return per unit of downside risk, offering a more realistic view of performance than total-risk measures.

You can estimate the Sortino ratio using R and explain the output.

This supports:

CLO2: Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.

and

CLO3: Use business analytics tools to develop, evaluate, and communicate predictive insights that support financial problem-solving.

You can now interpret advanced risk measures used in modern portfolio management and financial regulation.

Communicating Portfolio Performance Clearly

Throughout the module, you interpreted and communicated:

- Return measures
- Risk measures
- Risk-adjusted performance metrics
- Inflation-adjusted outcomes
- Downside-risk measures

You learned how to:

- Compare portfolios fairly.
- Explain why different metrics may produce different rankings.
- Connect quantitative measures to investment objectives and investor profiles.

This supports:

CLO4: Create and communicate financial insights using effective visual, written, and oral formats suited to both technical and non-technical stakeholders.

You can now explain portfolio performance in both technical and business language.

Practice Tasks

Practice Task 1: Holding Period Return

A portfolio is purchased for AUD 30,000 and sold one year later for AUD 32,400. During the year, dividends of AUD 600 are received.

Calculate:

- Holding Period Return (HPR)

Explain:

- What this return represents
 - Why HPR alone does not fully evaluate performance
-

Practice Task 2: Arithmetic vs. Geometric Returns

A portfolio earns:

- +12% in Year 1

- -5% in Year 2
- +8% in Year 3

Calculate:

- Arithmetic average return
- Geometric average return

Explain:

- Why the geometric average is lower
 - Why compounding matters for long-term investors
-

Practice Task 3: Risk-Adjusted Performance

Assume:

- Risk-free rate = 3%

Portfolio A:

- Return = 11%
- Standard deviation = 15%
- Beta = 1.2

Portfolio B:

- Return = 9%
- Standard deviation = 10%
- Beta = 0.7

Calculate:

- Sharpe Ratios
- Treynor Ratios

Explain:

- Which portfolio performs better on a risk-adjusted basis
 - Why Sharpe and Treynor may produce different rankings
-

Practice Task 4: Fisher Equation and Real Returns

A portfolio earns a nominal return of 8% while inflation is 5%.

Calculate:

- The real return using the Fisher equation

Explain:

- Why real returns are more important than nominal returns for long-term wealth preservation
-

Practice Task 5: Downside Risk

Suppose two hedge funds have similar Sharpe Ratios, but one has a much lower Modified Sharpe Ratio and higher CVaR.

Explain:

- What this suggests about tail risk
 - Why downside risk matters for portfolio management
 - Why investors may prefer downside-focused measures in volatile markets
-

Answers available [here \(https://learn.usc.edu.au/courses/35359/files/3294279?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3294279?wrap=1).

Reflections

Write short responses to the following:

- Why is it important to evaluate returns relative to risk?
 - Why can diversification reduce some risks but not others?
 - Why might a portfolio with lower returns still be superior on a risk-adjusted basis?
 - Why do investors increasingly use downside-risk measures such as Sortino Ratio?
 - Why are real returns more important than nominal returns during inflationary periods?
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Workshop

[Tasks \(https://learn.usc.edu.au/courses/35359/files/3282928?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3282928?wrap=1)